

ΞΕΙΛ Mathematics — Paper Diagnostic

Diagnostic paper · 30 questions · no calculator · 60 minutes

INSTRUCTIONS.

Answer every question. Choose ONE option and write its letter in the box on your answer sheet. There is no penalty for a wrong answer, so do not leave a question blank — but if you are guessing, say so on the answer sheet, because this paper is here to find out what to teach you, not to rank you.

ALGEBRA

1. Evaluate $-3 + 5 \times (-2)$.

A. 7

B. 13

C. -13

D. -4

2. Evaluate $\frac{2}{3} + \frac{1}{4}$.

A. $\frac{7}{12}$

B. $\frac{11}{12}$

C. $\frac{3}{7}$

D. $\frac{3}{4}$

3. Solve $5x - 7 = 3x + 9$.

A. $x = 8$

B. $x = 1$

C. $x = -8$

D. $x = 16$

4. Simplify $\frac{x^5 \cdot x^3}{x^4}$.

A. x^8

B. x^{12}

C. x^{11}

D. x^4

5. Factorise $x^2 - 7x + 12$.

A. $(x - 2)(x - 6)$

B. $(x - 1)(x - 12)$

C. $(x - 3)(x - 4)$

D. $(x + 3)(x + 4)$

6. Solve $\begin{cases} 2x + y = 11 \\ x - y = 1 \end{cases}$

A. (5, 1)

B. (4, 3)

C. (3, 4)

D. (4, -3)

7. If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$.

A. 9

B. 49

C. 19

D. 10

8. Solve $\log_2 x + \log_2(x - 2) = 3$.
- A. $x = 4$ or $x = -2$ B. $x = 5$
C. $x = -2$ D. $x = 4$
9. The equation $2x^2 - 6x + 1 = 0$ has roots x_1 and x_2 . Find $x_1 + x_2$.
- A. 3 B. $\frac{1}{2}$
C. 6 D. -3
10. Solve $x^2 - 5x + 6 < 0$.
- A. $2 \leq x \leq 3$ B. $x < 2$ or $x > 3$
C. $2 < x < 3$ D. $-3 < x < -2$

GEOMETRY & TRIGONOMETRY

11. Two parallel lines are cut by a transversal. One interior angle is 65° . Find the co-interior (allied) angle on the same side.
- A. 115° B. 25°
C. 65° D. 130°
12. A right-angled triangle has legs 5 and 12. Find the hypotenuse.
- A. 17 B. $\sqrt{119}$
C. 7 D. 13
13. Find the area of a circle of radius 6.
- A. 36π B. 36
C. 6π D. 12π
14. Two similar triangles have sides in the ratio 3 : 5. The smaller has area 18. Find the area of the larger.
- A. 45 B. 50
C. 30 D. 90
15. Evaluate $\sin 60^\circ \cdot \cos 30^\circ$.
- A. $\frac{3}{4}$ B. $\frac{\sqrt{3}}{2}$
C. $\frac{1}{4}$ D. $\frac{1}{2}$
16. Find the total surface area of a cuboid measuring $3 \times 4 \times 5$.
- A. 60 B. 47
C. 120 D. 94

17. A cone has base radius 3 and slant height 5. Find its volume.
- A. 36π B. 45π
C. 12π D. 15π
18. For $\vec{a} = (3, 4)$ and $\vec{b} = (2, -1)$, find $\vec{a} \cdot \vec{b}$.
- A. 7 B. 2
C. 10 D. -2

ANALYSIS

19. Differentiate $y = 3x^4 - 2x$.
- A. $3x^3 - 2$ B. $12x^3 - 2x$
C. $12x^3 - 2$ D. $12x^4 - 2$
20. Find $\int 6x^2 dx$.
- A. $12x + C$ B. $6x^3 + C$
C. $3x^3 + C$ D. $2x^3 + C$
21. Find the tangent to $y = x^2 - 3x$ at the point where $x = 2$.
- A. $y = -2x + 2$ B. $y = x - 2$
C. $y = x - 4$ D. $y = x + 4$
22. Evaluate $\int_0^2 3x^2 dx$.
- A. 12 B. 8
C. 6 D. 4
23. Find the local maximum VALUE of $f(x) = x^3 - 3x$.
- A. -1 B. -2
C. 2 D. 1

PROBABILITY & STATISTICS

24. Find the median of 9, 3, 14, 7, 7.
- A. 9
B. 7
C. 8
D. 14
25. A bag holds 3 red and 5 blue balls. One is drawn at random. Find $P(\text{red})$.
- A. $\frac{3}{8}$
B. $\frac{1}{3}$
C. $\frac{5}{8}$
D. $\frac{3}{5}$

26. Find the interquartile range of 2, 4, 5, 7, 8, 10, 13.
A. 11
B. 7
C. 5
D. 6
27. $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Find $P(A \cup B)$.
A. 0.7
B. 0.9
C. 0.2
D. 1.1
28. A random variable X takes the values 0, 1, 2 with probabilities 0.2, 0.5, 0.3. Find $E(X)$.
A. 0.5
B. 1.1
C. 1
D. 1.5

29. A student has 4 shirts and 3 pairs of trousers. How many different outfits are possible?

A. 12 B. 7
C. 24 D. 64

30. In how many ways can 3 students be chosen from 8?

A. 24 B. 112
C. 336 D. 56